BNW2T Series
Uncooled Infrared Thermal Module
Product Manual



Document Version: V1.2 (January 9, 2026)

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Change History

Changes between document issues are cumulative. The latest document issue contains all the changes in previous issues.

Version	Release Date	Revision Content
V1.0	October 23, 2025	Create
V1.1	December 17, 2025	1. Updated product selection description. 2. Updated 2D drawings.
V1.2	January 09, 2026	Corrected lens description for product selection.

1. Product Introduction

1.1. Product Description

The BNW2T series temperature measurement modules support 256/384/640 resolution across the entire lineup, featuring a self-developed high-frame-rate 12μm WLP packaged detector. Integrated with self-developed infrared ISP chip (second generation), these modules offer advantages including high performance, compact size, lightweight design, low power consumption,and cost-effectiveness, fully meeting SWaP (Size, Weight, Power/Price) application requirements.

Combined with the new-generation dedicated infrared image processing algorithms, the modules deliver finely detailed and clear image output. Additionally, the modules provide a variety of digital interfaces, enabling flexible integration with diverse intelligent processing platforms.

1.2. Product Features

- Powerful temperature measurement performance
- Standard configuration with shutter-less algorithm
- High-quality image output
- High cost-effectiveness

1.3. Application Scenarios

- Handheld temperature measuring device
- Security/fire inspection: industrial monitoring, perimeter scanning, inspection robots, fire warning, fire helmets.
- Intelligent integration: assisted driving, smart home appliances.

2. Product Selection

2.1. Model Encoding Rules

Table 2-1 BNW2T Series Model Coding Rules

Encoded Field	Field Description	Optional Codes/Examples
BNW2T	Product	BNW2T
384	Resolution	**256:** 256×192 **384:** 384×288 **640:** 640×512
09111	Lens (focal length, F-number)	**04011:** 4.0mm F1.0 **06812:** 6.8 mm F1.2 **09111:** 9.1 mm F1.1 **13510:** 13.5 mm F1.0 **18011:** 18 mm F1.1 For more lens codes, please refer to "2.3 Lens Optical Specifications".
A	Lens Serial Number	**A:** The first lens for the focal length, F-number and focus method. **B:** The second lens.
S	Gain Mode	**S**: Two level gain
S	Focus Type	**S:** Fixed focus **M:** Manual focus
W	Operating Modes	**W:** Temperature measurement **N:** Imaging
C	Expansion Board	**C:** No USB

NOTE 1. This series of modules supports the optional configuration without lens.

2.2. Available Models List

Table 2-2 BNW2T 256 Module Optional Model List

Lens		BNW2T 256 Model Selection	FOV
Focal Length (mm)	F-number		
4	1.1	BNW2T256T04010ASSWC	45.5°×33.5°

Table 2-3 BNW2T 384 Module Optional Model List

Lens		BNW2T 384 Model Selection	FOV
Focal Length (mm)	F-number		
4	1.1	BNW2T38404011ASSWC	64.1°×48.8°
6.8	1.2	BNW2T38406812ASSWC	39.3°×29.4°
9.1	1.1	BNW2T38409111ASSWC	29.2°×21.7°

Table 2-4 BNW2T 640 Module Optional Model List

Lens		BNW2T 640 Model Selection	FOV
Focal Length (mm)	F-number		
9.1	1.1	BNW2T64009111ASSWC	48.7°×38.6°
13.5	1.0	BNW2T64013510ASSWC	31.9° × 25.7°
18	1.1	BNW2T64018011ASSWC	24.2°×19.5°

NOTE 1. The above are all the first lenses (A), and will be updated if there are any changes in the future.

NOTE 2. The symbol "×" indicates that the model is not supported.

NOTE 3. If necessary, you can contact our company to customize the selection combination that is not recommended in the list.

2.3. Lens Optical Specifications

Table 2-5 Lens Optical Specifications and Weight

Lens Code	Lens Parameters			Focus (mm)	Weight (incl. flange) / g
	Focal Length (mm)	F-number	IFOV (mrad)		
04011	4	1.1	3.05	0.29~∞	10±5%
06812	6.8	1.2	1.76	0.8~∞	9.3±5%
09111	9.1	1.1	1.3	1.2~5.5	10.5±5%
13510	13.5	1.0	0.9	3.8~∞	15.2±5%
18011	18	1.1	0.7	6.14~∞	25±5%

NOTE 1. The weight here only indicates the weight of the lens. The whole weight includes lens, module, FPC, optional USB accessory.

NOTE 2. There is no FOV in the above table, please refer to "2.2 Available Models List" in this manual for this parameter.

NOTE 3. Focus refers to the maximum clear range that can be obtained when fixed focusing (dispensing or fixed using a headless set screw), not the focusing range, which is typically 0.3 m~∞.

Table 2-6 Range Performance

Lens Code	Detection Distance (km)	Recognition Distance (km)	Identification Distance (km)
04011	0.3	0.1	0.05
06812	0.51	0.17	0.09
09111	0.68	0.23	0.11
13510	1.5	0.37	0.19
18011	2	0.5	0.25

NOTE 1. The above detection distance, recognition distance and identification distance are estimated according to Johnson's Criteria, taking pedestrians (2×0.5×0.3 m) as the target.

3. Specifications

Table 3-1 BNW2T Series Technical Specifications

Overview	BNW2T		
Detector Type	Vanadium Oxide Uncooled Infrared Focal Plane Detector		
Resolution	256×192	384×288	640×512
Effective Frame Rate	25Hz	30Hz	
Pixel Pitch	12μm		
Spectral Band	8～14μm		
Noise Equivalent Temperature Difference (NETD)	≤40mK@25℃,F#1.0,25Hz		
Thermal Time Constant	＜12ms		
Image Adjustment			
Non-uniformity Correction	Shutter correction/shutter-less algorithm correction		
Brightness/Contrast	0~100, optional		
Polarity/Palette	White-hot/Black-hot/Various palettes supported		
Digital zoom	1.0~8.0×continuous zoom (step size: 0.1)		
Image Mirroring	Vertical/Horizontal/Diagonal		
Electricity Parameters			
Digital Video	DVP/ USB2.0/MIPI		
Communication Interface	UART/I2C/USB2.0		
Power Supply Input	5V		
Typical power consumption of USB output @25℃	＜0.35W	＜0.42W	＜0.7W
Physical Characteristics			
Weight (without lens)	<7g		<8.6g
Dimensions (without lens)	21×21×10.3 mm		
Temperature Measurement			
Measurement Range	-20℃~150℃ (high-quality), 100℃~650℃ (wide range)		
Measurement Accuracy	±2°C or ±2% (typical value)		
Environment Adaptability			
Operating Temperature	-40℃ ~ +80℃ (imaging), -10℃ ~ +70℃ (temperature measurement)		
Storage Temperature	-50℃～+85℃		
Humidity	5%–95%, non-condensing		
Vibration	6.06g, random vibration, all axes		
Impact	80g, 4ms, final peak sawtooth wave, three axes and six directions		

NOTE 1. The above parameters are measured under laboratory conditions, the actual specifications and test methods are subject to the test report.

4. Interfaces and Accessories

4.1. Interface Introduction

50PIN male connector model: DF40C-50DP-0.4V (51)

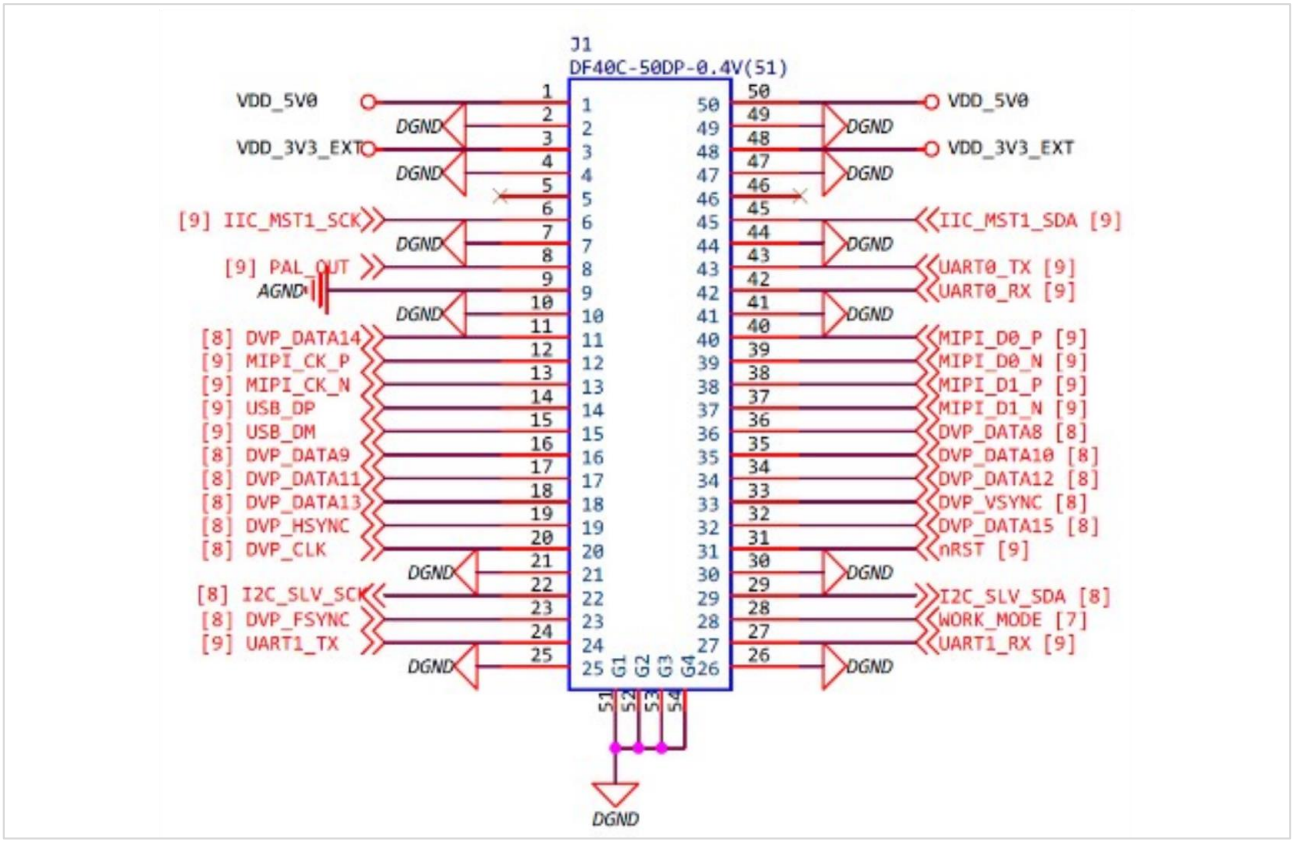


Figure 4-1 Hirose50 Pinout Diagram A

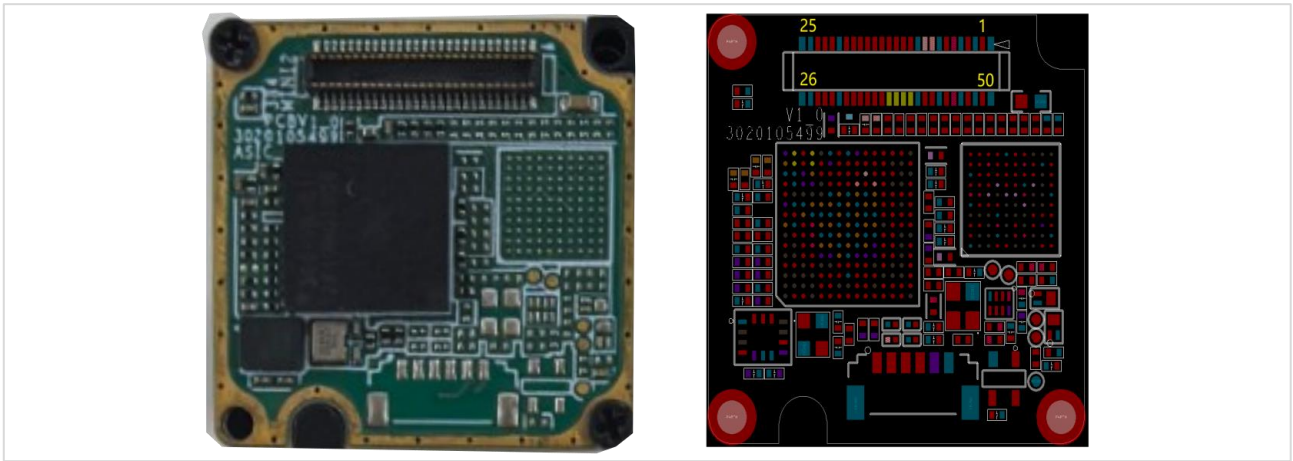


Figure 4-2 Hirose50 Pinout Diagram B

Table4-1 HIROSE50 PIN Definition

No.	Name	Type	Revision Content
20, 33, 19, 32, 11, 18, 34, 17, 35, 16, 36	DVP_CLK DVP_VSYNC DVP_HSYNC DVP_DATA15~8	Output	DVP data signal DATA15: MSB DATA8: LSB
12, 13, 40, 39, 38, 37	MIPI_CK_P/N MIPI_D0_P/N MIPI_D1_P/N	Output	2lane MIPI data signal
14, 15	USB_DP USB_DM	Output	USB2.0 Data signal
8, 9	PAL_OUT AGND	Output	Analog output signal Analog output ground
43, 42	UART0_TX UART0_RX	Output	Universal asynchronous receiver/transmitter pins for control module use.
24, 27	UART1_TX UART1_RX	I/O	Universal asynchronous receiver/transmitter pins for debugging, which can print the module's operating logo. They can also be used as general I/O.
22, 29	I2C_SLV_SCK I2C_SLV_SDA	I/O	Clock pin of IIC slave Data pin of IIC slave
6, 45	IIC_MST1_SCK IIC_MST1_SDA	I/O	Clock pin of IIC master Data pin of IIC master
23	DVP_FSYNC	I/O	It is an external synchronous input signal of the module, not a DVP synchronous output signal, and can also be used as a general I/O.
28	WORK_MODE	Input	Enter to update mode; Active high; Leave floating if unused.
31	Nrst	Input	Enter to reset mode; Active low; Leave floating.
1, 50	VDD5.0	Power	External power input 5 V, and the module can operate normally with only 5 V power supply.
3, 48	VDD3.3	Power	External power input 3.3 V, it is only reserved and does not require power supply.
2, 4, 7, 10, 21, 25, 26, 30, 41, 44, 47, 49	DGND	Power	GROUND
5, 46	NC	NC	Not available

NOTE 1. The order of pin numbers and pin names in the table is consistent.

NOTE 2. The default voltage for DVP and IIC slave is 1.8 V, which can be switched to 3.3 V by command.

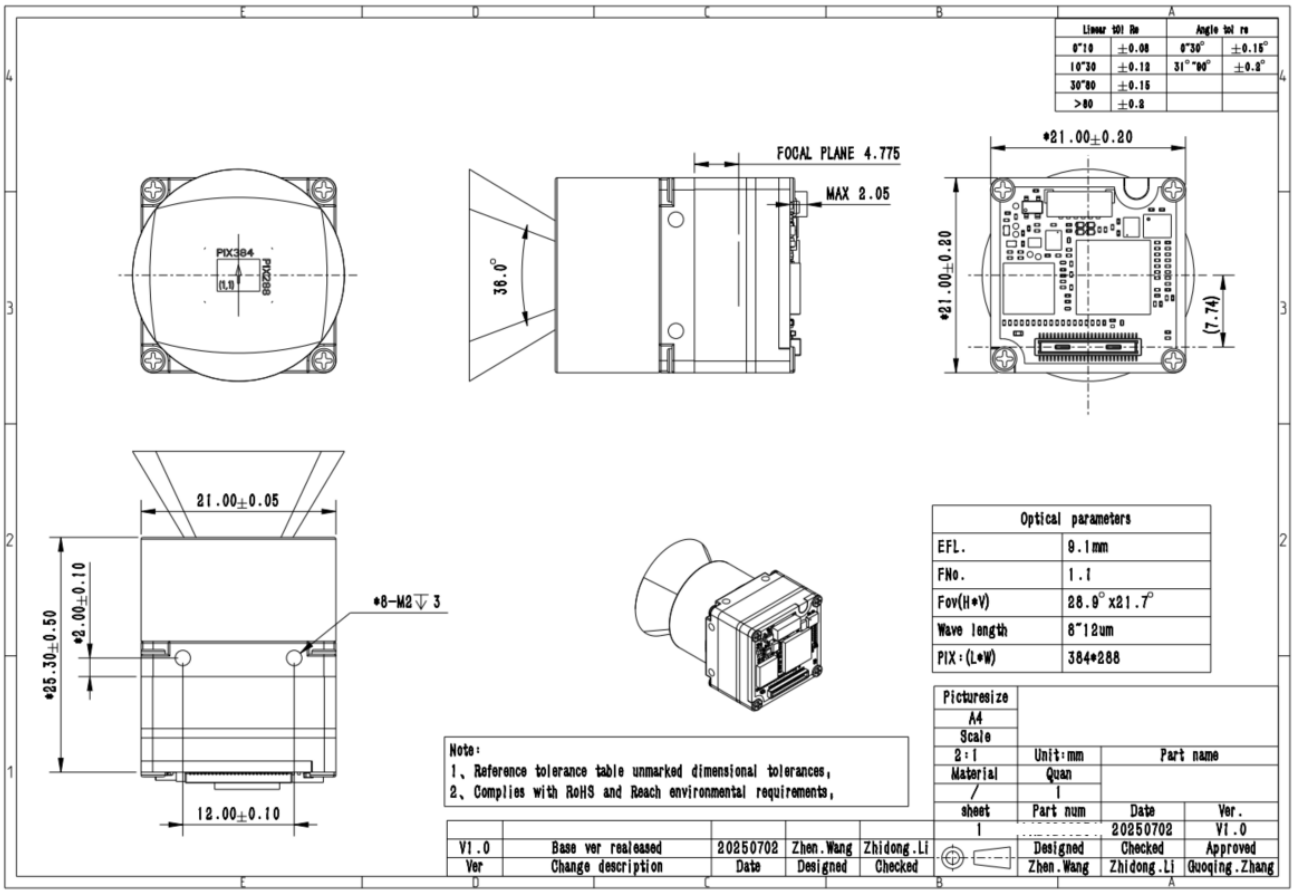
NOTE 3. The default voltage for UART0, UART1, and IIC master is 3.3V and cannot be switched.

NOTE 4. The hardware description here is only for product selection reference. Please contact our technicians for detailed hardware development information during specific development.

NOTE 5. CVBS only supports black-hot/white-hot switching.

5. Structure and Dimensions

This chapter takes the drawing of the BNW2T 384 9.1mm lens module as an example to illustrate its typical structural information. For detailed 2D/3D engineering drawings of this model or other lens modules, please contact our technical support department.



6. Precautions

To protect you and others from injury or to protect your equipment from damage, please read all the information below before using the equipment.

1. Do not look directly at high-intensity radiation sources such as the sun.
2. The ideal operating environment temperature is -40℃~80℃.
3. Do not touch devices and cables with wet hands.
4. Do not bend or damage the cables.
5. Do not scrub your device with diluent.
6. Do not unplug other cables before cutting off the power supply.
7. Do not connect the attached cables in the wrong way to avoid damaging the device.
8. Take measures to prevent static electricity.
9. Do not dismantle the device. If there is any fault, please contact us to have it repaired by professionals.

7. Warranty

Dear User,

Thank you for choosing our products, we will, as always, to provide you with satisfactory service!

1. This product in normal use under the circumstances of failure the company will provide 1-year warranty, life-long maintenance services.
2. Warranty scope:

   Failure under normal circumstances is generally defined as natural damage caused by the user of the product during normal use without human intent or due to negligence factors.
3. The following cases are not in our warranty scope:
   1) Any damage caused by modification or repair not authorized and permitted by the company.
   2) Failure or damage caused by the use of third-party product software, service behavior.
   3) Accidental factors or human behavior caused by product damage. Such as into the liquid, drop damage, input the wrong voltage, excessive extrusion, deformation of the motherboard and so on. Appearance of obvious hard object damage, cracks, broken foot, serious deformation, power cord broken, broken cable, bare core and other phenomena.
   4) Product data loss or damage.
   5) Cannot effectively present the product warranty certificate. (Product nameplates, SN barcodes, and tamper-evident labels are torn off or damaged, blurred and unrecognizable.)
   6) Not in accordance with the instructions for installation, use, maintenance, storage of the product failure or damage.
   7) It has exceeded the warranty period.
   8) Failure or damage due to uncontrollable factors (e.g. fire, earthquake, flood, etc.).

8. Supports and Services

The company provides services such as pre-sales training, in-sales support, and after-sales maintenance. Please contact our sales-reps for specific information.